

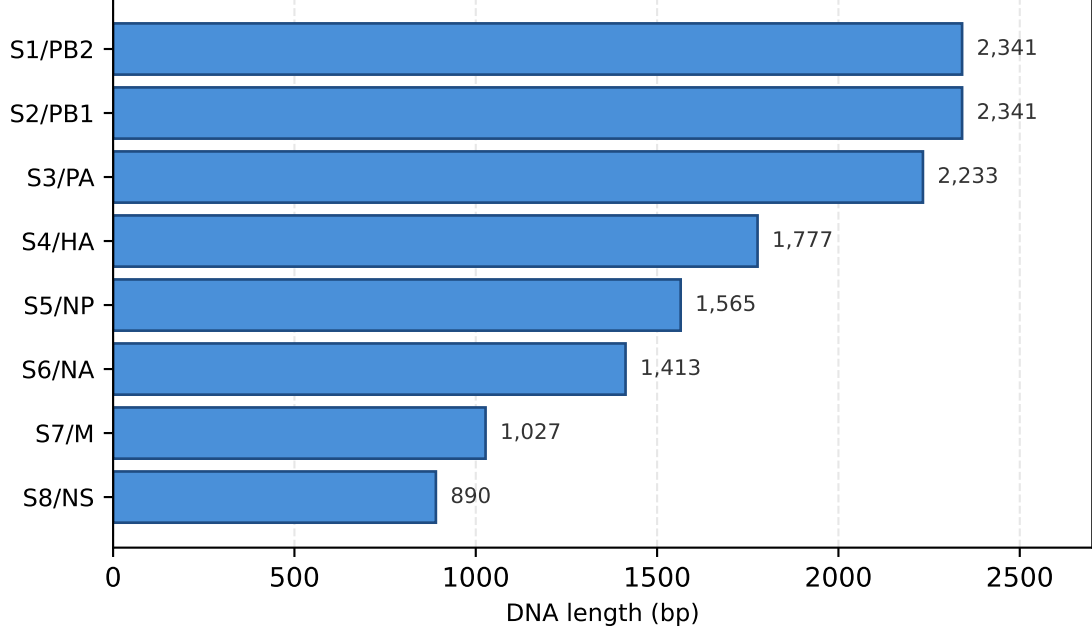
K-mer feature pipeline: GTO contigs → sparse k-mer matrix → pair features for MLP

A. Input: GTO (Genome Typing Object) file

```
{
  "scientific_name": "Influenza A virus (A/.../H3N2)",
  "ncbi_taxonomy_id": 197911,
  "quality": { "genome_quality": "Good" },
  "contigs": [
    { "id": "1406633.10", "replicon_type": "Segment 1",
      "dna": "tcaaatatattcaatatgga...", "length": 2341 },
    { "id": "1406633.8", "replicon_type": "Segment 2",
      "dna": "caaaccatttgaatggatgt...", "length": 2341 },
    ... (6 more segments: PA, HA, NP, NA, M, NS) ...
  ],
  "features": [ {"type": "CDS", "protein_translation": "...", ...} ]
}
```

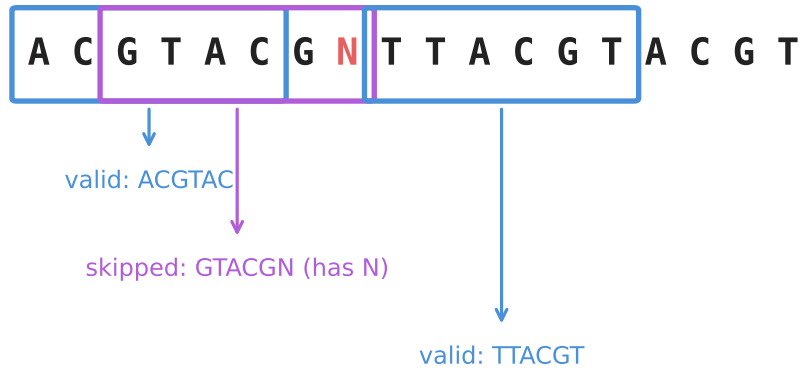
One GTO per isolate. `contigs[]` has one entry per genomic segment (DNA).
`features[]` has one entry per CDS (protein). Stage 1 extracts both.

B. Stage 1: genome_final.csv — one row per (assembly, segment)



Aggregated across all isolates → 868,240 contig rows.
Columns: assembly_id, genbank_ctg_id, canonical_segment (S1..S8), dna_seq, length, gc_content.

C. Stage 2b: slide a k=6 window along each DNA sequence

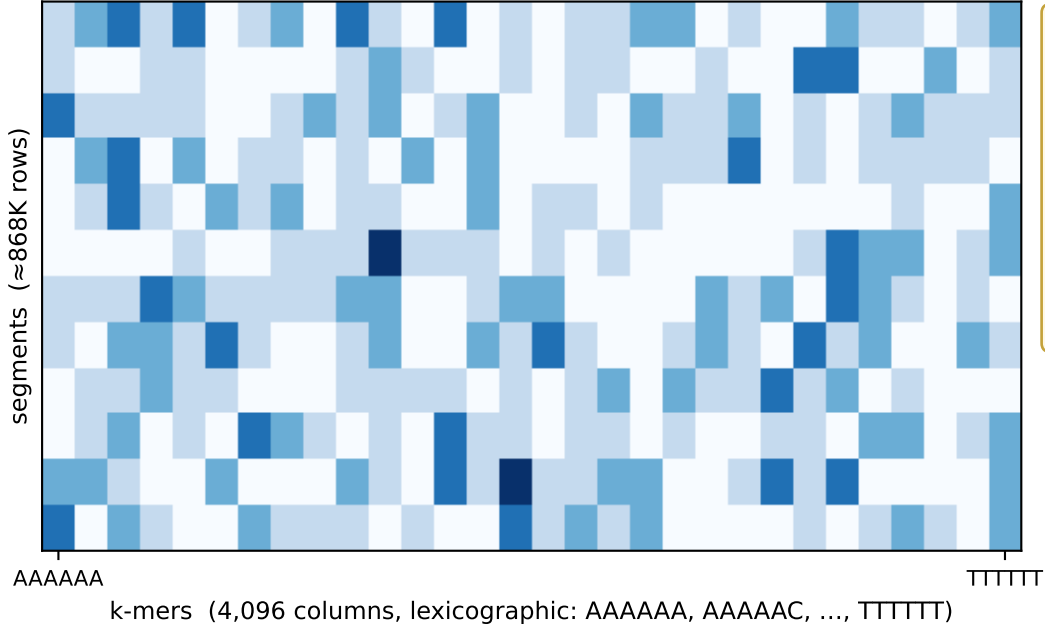


Rules:

- Step = 1 (all overlapping windows).
- Alphabet = {A, C, G, T}; windows containing any ambiguous base (N, R, Y, ...) are skipped.
- No reverse-complement canonicalization; no strand awareness.

Result per sequence: an integer count for each of the 4⁶ = 4,096 possible 6-mers.

D. Stacked sparse k-mer matrix: one row per segment, 4,096 columns



Real shape

rows (segments): 868,240

cols (6-mers): 4,096

non-zeros: 1,049,945,579

sparsity: 70.5% zero

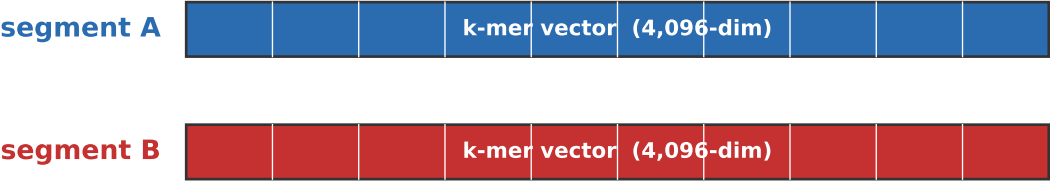
avg distinct 6-mers per segment: ≈ 1,209

Storage

scipy.sparse CSR (.npz)

+ parquet index mapping (assembly_id, genbank_ctg_id, canonical_segment) → row

E. Pair feature for the classifier: lookup → interaction → MLP



Lookup key:
(assembly_id, genbank_ctg_id)
Pair CSV → index parquet → matrix row

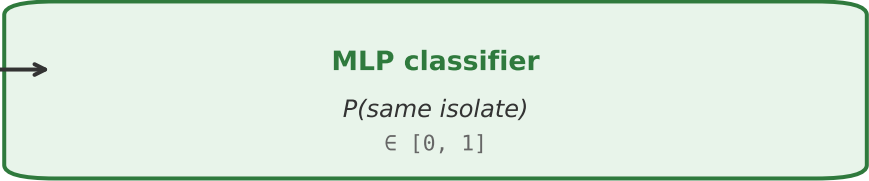
Interaction f(A, B):

concat: [A ; B] → 8,192-dim (used in this paper's 28-pair sweeps)

diff: |A - B| → 4,096-dim

unit_diff: (A - B) / ||A - B|| → 4,096-dim

prod: A ⊙ B → 4,096-dim



Note: multiple proteins can share one contig (M1/M2, NS1/NEP, PA/PA-X) → identical k-mer vector for both. Paper's 28-pair sweeps avoid this by selecting one function per segment.